sexM/sexP candidate HMG-box tree, FastTree -lg -gamma (hmmalign PF00505, 69 columns); labels from classifier scan 1a00b0a

clade
🟦 curated sexM reference
🟥 curated sexP reference
🟪 Ascomycota MAT1-2-1 (root: F. graminearum)
🟦 detection label Minus
🟧 detection label Plus
⬛ no label (non-locus copy / undetermined)
strip: best sexM group (7 of 9 refs; local support 0.76; not monophyletic)
strip: sexP clade (all 6 refs; local support 0.23)
● FastTree local support >= 0.95
○ FastTree local support 0.80-0.94
Mucor racemosus CGMCC-40320 (withheld)
Mucor racemosus C (withheld)
Parasitella parasitica CBS 412.66 substr. NGI-315 ade (withheld)
Mucor lusitanicus CBS 277.49 (withheld)
Mucor griseocyanus CBS 116.08 (withheld)
Mucor sp. M7501 M7501 (withheld)
Mucor fragilis (withheld)
Mucor janssenii CBS 205.68 (withheld)
Mucor ambiguus NBRC 6742 (withheld)
Mucor lusitanicus MU402 (withheld)
Mucor ardhlaengiktus CBS 210.80 (withheld)
Thamnidium elegans KKP 4289 (non-locus copy)
Helicostylum pulchrum RSA 2064 (non-locus copy)
Pilaira anomala RSA1997 (non-locus copy)
Actinomucor elegans CJ-6 (withheld)
Mucor irregularis B50 (withheld) x2
Rhizopus arrhizus GL4 (withheld) x6
Rhizopus arrhizus HUMC 02 (withheld)
Mycotypha africana NRRL 2978 (withheld)
Mucor circinelloides B8987 (withheld)
Mucor circinelloides WJ11 (withheld)
Mucor circinelloides CBS 394.68 (withheld) x3
Mucor circinelloides CBS 195.68 (withheld)
Mucor circinelloides 1006PhL (withheld) x9
Spinellus fusiger NRRL 22323 (non-locus copy)
Syncephalastrum racemosum NRRL 2496 (withheld)
Lichtheimia sp. PT1902 PT1902 (withheld)
Lichtheimia ornata CBS 291.66 (withheld)
Lichtheimia ramosa B5792 (withheld) x7
Lichtheimia ramosa B5399 (withheld) x2
Lichtheimia ramosa PG115-04A (withheld)
Lichtheimia corymbifera 008-049 (withheld) x5
Fennellomyces sp. T-0311 T-0311 (withheld)
Syncephalastrum sp. PYS2703 PYS2703 (withheld) x2
Syncephalastrum contaminatum 142427 (withheld)
Absidia sp. UoMAU379 UoMAU379 (withheld)
Absidia cylindrospora (withheld)
Absidia sp. YW-2025a CGMCC 3.27810 (withheld)
Absidia glauca CBS 101.48 substr. RVII-324 met (withheld)
Thermomucor indicae-seudaticae HACC 243 (withheld)
Rhizomucor miehei CAU432 (withheld)
Mucor sp. PG5414 PG5414 (withheld)
Backusella macrospora UoMAU7 (withheld)
Radiomyces spectabilis NRRL 2753 (withheld)
Mortierella alpina CGMCC 20262 (withheld)
Mortierella alpina GBA31 (withheld)
Mortierella alpina B6842 (withheld)
Mortierella alpina AD072 (withheld) x3
Mortierella antarctica UFMGCB 3722 (withheld)
Mortierella alpina CCTCC M207067 (withheld) x3
Mortierella alpina LL118 (withheld)
Mortierella alpina CK1249 (withheld)
Entomortierella beljakovae KOD1040 (withheld)
Mortierella sp. AD010 AD010 (withheld) x2
Mortierella sp. AM989 AM989 (withheld)
Haplosporangium sp. Z 27 Z 27 (withheld)
Mortierella sp. BCC40632 BCC40632 (withheld)
Lunasporangiospora selenospora KOD1015 (withheld)
Dissophora ornata Z 71 (withheld)
Gamsiella multidivaricata NRRL 6456 (withheld) x2
Modicella reniformis MES-2147 (withheld)
Rhizopus arrhizus {'name': 'CBS 346.36', 'known': True, 'culture_collection_ids': ['CBS 346.36'], 'differs_from_sequenced': False} [REF sexP]
Syzygites megalocarpus {'name': 'ATCC 11807', 'known': True, 'culture_collection_ids': ['ATCC 11807'], 'differs_from_sequenced': False} [REF sexP]
Mucoraceae sp. AI-2023a {'name': 'AI-2023a', 'known': True, 'culture_collection_ids': ['AI-2023a'], 'differs_from_sequenced': False} [REF sexP]
Mucor circinelloides {'name': 'NRRL 3631', 'known': True, 'culture_collection_ids': ['NRRL 3631'], 'differs_from_sequenced': False} [REF sexP]
Phycomyces blakesleeanus {'name': 'UBC21', 'known': True, 'culture_collection_ids': ['UBC21'], 'differs_from_sequenced': False} [REF sexP]
Absidia urquhartii {'name': 'UoMD23-19', 'known': True, 'culture_collection_ids': ['UoMD23-19'], 'differs_from_sequenced': False} [REF sexP]
Tuber indicum {'name': 'ysh1-8', 'known': True, 'culture_collection_ids': ['ysh1-8'], 'differs_from_sequenced': True} [REF MAT1-2-1]
Rhizopus azygosporus {'name': 'CBS 357.93', 'known': True, 'culture_collection_ids': ['CBS 357.93'], 'differs_from_sequenced': False} [REF sexM]
Syzygites megalocarpus {'name': 'ATCC 11807', 'known': True, 'culture_collection_ids': ['ATCC 11807'], 'differs_from_sequenced': False} [REF sexM]
Rhizopus arrhizus {'name': 'CBS 110.17', 'known': True, 'culture_collection_ids': ['CBS 110.17'], 'differs_from_sequenced': False} [REF sexM]
Mucor mucedo {'name': 'CBS 109.16', 'known': True, 'culture_collection_ids': ['CBS 109.16'], 'differs_from_sequenced': False} [REF sexM]
Mucor hiemalis
Mucor irregularis B50 x2
Cunninghamella elegans B9769
Cunninghamella bainieri 2A1
Cunninghamella echinulata
Gongronella sp. w5 w5
Gongronella butleri CBS 227.36
Absidia glauca CBS 101.48 substr. RVII-324 met
Absidia urquhartii {'name': 'UoMAU377', 'known': True, 'culture_collection_ids': ['UoMAU377'], 'differs_from_sequenced': False} [REF sexM]
Absidia healeyae UoMAU1
Absidia sp. YW-2025a CGMCC 3.27810
Absidia cylindrospora
Halteromyces radiatus CBS 162.75
Endogone sp. FLAS-F59071 FLAS-F59071 (non-locus copy)
Umbelopsis ramanniana (non-locus copy)
Bifiguratus adelaidae AZ0501 (non-locus copy)
Spiromyces aspiralis RSA 2271 (non-locus copy)
Mycoemilia scoparia NBRC 100468 (withheld)
Mortierella sp. GBA35 GBA35 (withheld)
Mortierella sp. NVP41 NVP41 (withheld)
Mortierella sp. AD031 AD031 (withheld)
Entomortierella parvispora E1425 (non-locus copy)
Lobosporangium transversale NRRL 3116 (non-locus copy)
Benniella sp. AD185 (non-locus copy)
Dissophora ornata Z 71 (non-locus copy)
Gryganskiella cystojenkinii 1230 (non-locus copy)
Gamsiella multidivaricata RSA 2152 (non-locus copy)
Podila epicladia KOD1059 (withheld)
Actinomortierella ambigua BC1291 (non-locus copy)
Actinomortierella ambigua BC1291 (non-locus copy)
Podila verticillata LANL.1351.123 (non-locus copy)
Podila verticillata LANL.1351.123 (non-locus copy)
Mortierella sp. AD010 AD010 (withheld) x2
Entomortierella beljakovae KOD1040 (withheld)
Haplosporangium sp. Z 27 Z 27 (non-locus copy)
Linnemannia zychae PUST F9C (withheld)
Linnemannia fluviae (non-locus copy)
Linnemannia hyalina SCG-10 (withheld)
Linnemannia elongata GBA34 (withheld)
Linnemannia schmuckeri NRRL 6426 (withheld)
Linnemannia fluviae (non-locus copy)
Mortierella sp. AD031 AD031 (withheld)
Mortierella sp. NVP41 NVP41 (withheld)
Mortierella sp. GBA35 GBA35 (withheld)
Mortierella sp. GBA30 GBA30 (withheld)
Gryganskiella cystojenkinii 1230 (non-locus copy)
Entomortierella parvispora E1425 (non-locus copy)
Gamsiella multidivaricata RSA 2152 (non-locus copy)
Dissophora ornata Z 71 (non-locus copy)
Haplosporangium sp. Z 27 Z 27 (non-locus copy)
Benniella sp. AD185 (non-locus copy)
Modicella reniformis MES-2147 (non-locus copy)
Fusarium graminearum {'name': '3639', 'known': True, 'culture_collection_ids': ['3639'], 'differs_from_sequenced': True} [REF MAT1-2-1]
0.5 substitutions/site